

Accountability Gap

GLOSSARY 06 - SHEET 6 / 8

The condition in which harm occurs and no party can properly be held responsible

PROFESSIONAL DEFINITION

The accountability gap - also termed the responsibility gap - is the condition in which an autonomous or learning system causes harm and no human party satisfies the conditions ordinarily required for moral or legal responsibility, because no one had the control or the foreseeability that responsibility presupposes. Many-hands problems in distributed development compound it, as responsibility diffuses across designers, deployers, operators and data providers (Matthias, 2004; Santoni de Sio and Mecacci, 2021).

WHO DEFINED IT ORIGINALLY

Andreas Matthias identified and named the problem in Ethics and Information Technology in 2004. Filippo Santoni de Sio and Giulio Mecacci (2021) decomposed it into four distinct gaps - culpability, moral accountability, public accountability and active responsibility - in Philosophy and Technology. Helen Nissenbaum's (1996) work on many hands in computing is the direct antecedent.

WHY THIS DEFINITION WAS DEVELOPED

Traditional responsibility ascription requires that an agent could have known and could have done otherwise. Systems that learn after deployment weaken both conditions, so harms could occur for which no one was answerable. Naming the gap made it a design and governance problem - to be closed by allocation - rather than an unavoidable feature of the technology.

HOW IT IS USED

Assign a named accountable person for each deployed system before deployment, ensure that person has both the information and the authority to act, and record the assignment; a gap identified after harm has occurred cannot be closed retrospectively.

WHEN IT IS USED

Before deployment of any autonomous or consequential system, at incident review, in vendor contracting where the model is supplied by a third party, and in insurance and liability assessment.

BY WHOM IT IS USED

Boards accepting residual risk, general counsel and risk officers, system owners named as accountable, and regulators determining where liability rests.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position was product liability, which attaches to a defect in an artefact with fixed properties at the point of sale. A system that adapts after deployment may cause harm without any defect at the point of supply, which product liability was not constructed to address. Santoni de Sio and Mecacci's four-way decomposition supplies the practical advance: the four gaps close by different means - contractual allocation, organisational design, transparency duties and meaningful human control - so the remedy depends on which gap is actually open.

RELATED DEFINITIONS IN THIS COURSE

AI Governance - Unit 6

Human-in-the-Loop - Unit 2

Explainability - Unit 6

Decision Rights - Unit 2

EU AI Act - Unit 6

SOURCES (HARVARD REFERENCING STYLE)

- Matthias, A. (2004) 'The responsibility gap: ascribing responsibility for the actions of learning automata', Ethics and Information Technology, 6(3), pp. 175-183.
- Nissenbaum, H. (1996) 'Accountability in a computerized society', Science and Engineering Ethics, 2(1), pp. 25-42.
- Santoni de Sio, F. and Mecacci, G. (2021) 'Four responsibility gaps with artificial intelligence: why they matter and how to address them', Philosophy and Technology, 34(4), pp. 1057-1084.

FIGURE 1 - FOUR GAPS, FOUR REMEDIES



Figure 1. The four gaps are distinct and close by different means. Naming an accountable owner addresses the second and does nothing for the fourth, which requires that the named person hold both the information and the authority to intervene (after Santoni de Sio and Mecacci, 2021).